Lab: 3D Basics in Adobe After Effects

One thing that all of the labs and assignments we've done in the course have in common is that they are all presented in a flat two-dimension setup. In this lab we will dive into the world of 3D within Adobe After Effects.

Unlike programs such as Adobe Photoshop and Adobe Illustrator that builds graphics in 2D space of length and width, 3D programs allow you to build compositions in 3D space, length, width and depth. There are 3D graphic programs such as Maya, Blender, and CAD. After Effects is what is referred to as a 2.5D program, in that you can place items in 3D space, but the objects themselves are flat or 2D.

In this lab we will be creating an animated 3D title intro for a nature documentary about animals in the Sergenti. This lab will discuss working with 3D layers, placing items in 3D space, building and animating cameras and working with lights in Adobe After Effects.

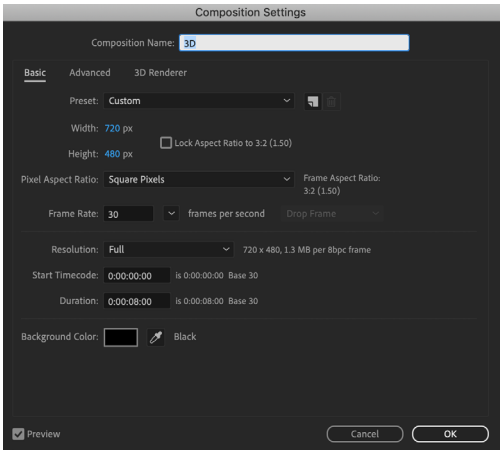
Setup



Take a moment to look at the video will will create using Adobe After Effects. You can view the video on YouTube below:

https://youtu.be/c-fSDK5c5CE

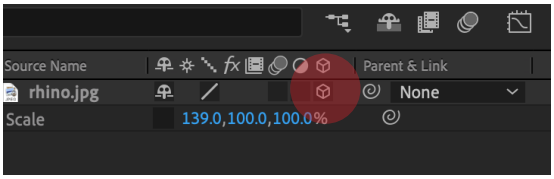
Make sure you have downloaded the lab files needed to complete this lab. The files are in zip folder called *Lab-3D-Basics.zip*. Make sure to unzip the folder completely to use the files inside appropiately.



Create a new After Effects project and import all of the images from the lab folder into our project. Create a new composition called *3D* is *720px x 480px*, frame rate of 30 frames / second and 8 seconds long.

Our client wanted us to build an interesting intro for their documentary of the "Animals of the Sergenti." They gave up several photos to work from, but no real video clips. Photos are fun, but can be a bit bland. We will place the animals in 3D space and build a camera to move through the images to create an intriguing faux 3D space for our intro.

3D Basics

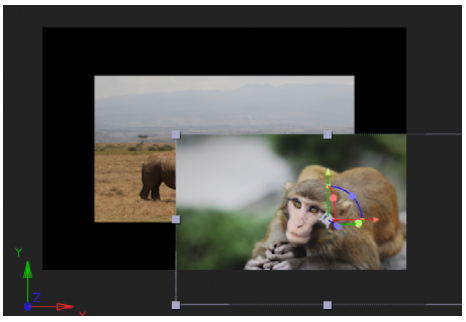


Click and drag the *Rhino.jpg* into the timeline. In the *Toggle Switch* area of the timeline click on the *3D Layer* checkbox, this will turn *Rhino.jpg* into a 3D layer. When a layer becomes 3D it can be positioned, rotated and scaled in 3D space.

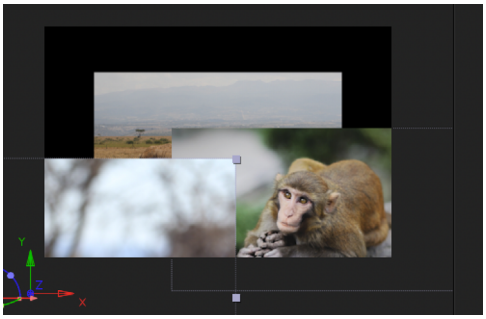
Highlight the *Rhino.jpg* layer and press the *"p"* key. You'll notice that now we have three values, instead of two. There is the ***x position***, *y-position* and now the *z-position* where the object exists in the depth of the scene. Think about our composition like a diarama for school. The *z-position* represents how close or far the layer is from the user or camera.



Set the *z-position* to *400*. You'll notice that the *Rhino.jpg* appears smaller than before. This is not because we scaled it smaller, but rather that is farther back is space from the audience than it was before. When objects are further back in 3D space they appear smaller than they really are, such as mountains along the horizon.



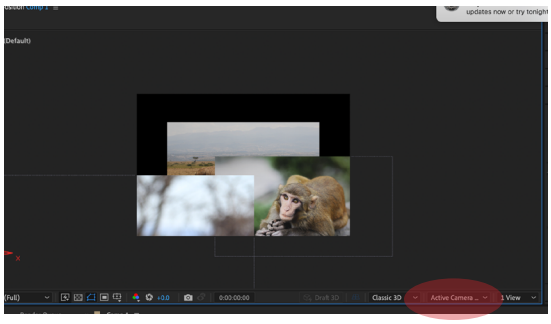
Drag *Monkey.jpg* into the timeline and make it into a 3D layer. Set the *z-position* to *200* and use the *Selection* tool to move the *Monkey.jpg* to the lower right hand corner as shown in the image on the right. You'll notice that the *Monkey.jpg* image appears larger, even though we can see in the *Project* panel that both *Rhino.jpg* and *Monkey.jpg* have the same dimensions. The reason why *Rhino.jpg* appears smaller is it further back in distance from the camera than *Monkey.jpg*, thus it appears smaller.



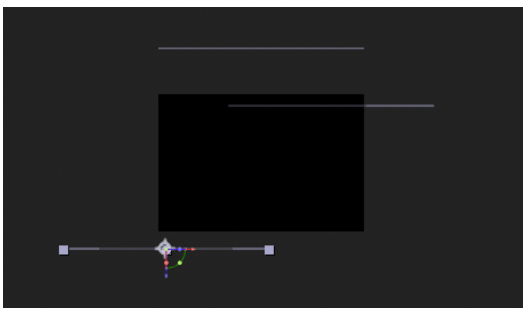
You can also set the *z-position* to a negative value as well. A negative value will bring the layer closer to the audience or camera. Select and drag *Gazelle.jpg* into the timeline. Make it a 3D layer and set it to *-300.* Use the Selection tool to move the image to the lower left hand corner and you'll see that image appears much closer to the audience / camera.

3D View Popup

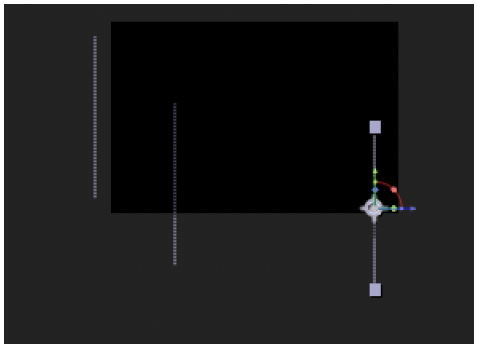
Now that we are dealing with 3D space it can be a bit confusing about where things exist. Is something small because it is scaled down, or because it further back in 3D space? To help with this Adobe After Effects has the *3D View Popup* menu. This allows us to view our scene in Adobe After Effects in orthographic views. To understand orthographic views imagine our scene is a diarama that we built. Then image we have multiple cameras that are positions on the same diarama, but from different angles and perspectives. These perspectives include the top, the bottom, the back, the front, left and right. These orthographic views give us multiple perspectives of the same After Effects scene.



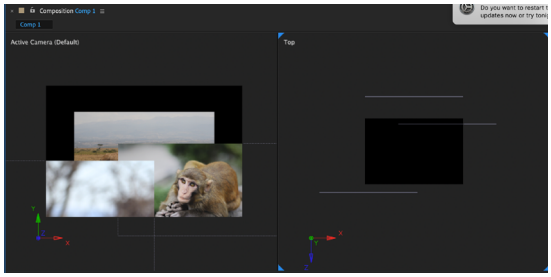
Begin by going to *Composition* panel and clicking on the *3D View Popup* on the bottom right, you'll see it is currently set to *"Active View"* or how the camera currently sees the scene. Set it to "*top"* or top orthographic. You see that the scene suddenly changes and instead of seeing a rhino and monkey you see two horizontal lines. You may need to zoom out to see the two horizontal lines. What are these? So you are looking at the scene from the top



view. These lines are actually your *Rhino.jpg* and *Monkey.jpg*. From the top, since both are just flat images look like lines. They are currently positioned facing the bottom, or where the camera is. Click on the *3D View Popup* and switch the view to left. Now instead of two horizontal lines you see two vertical lines, you may need to zoom / pan to see the two horizontal lines. The two vertical lines you are seeing the *Rhino.jpg* and *Monkey.jpg* from the left side. Think of it as view flats for a theatre show from the side. They are drawn, but have no depth and thus appear as thin lines.

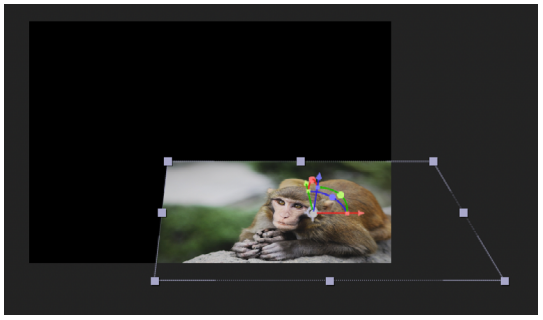


To return back to our usual view of our project set the *3D View Popup* back to *Active View*. To help view a complicated 3D scene we have multiple shots of the same scene in our *Composition* window. On the far right there is a dropdown called *Select View Layout*. Click and select *2 Views*, you'll see that this sets up two viewport in your composition window. Each viewport can be set to a different perspective. Select the viewport on the left, you can tell if it selected as it will have blue triangles around the edges. Set the left viewport to *"Active Camera."* Select the right viewport and set it to the *"top"* perspective.

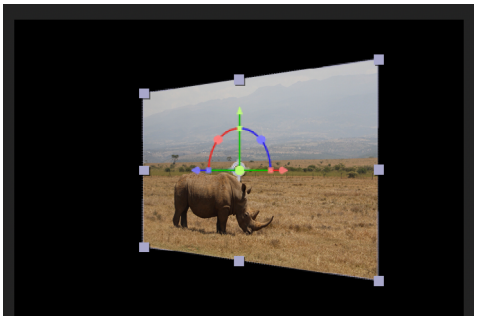


Rotation

Now that we have a basic undersanding of placing layers in 3D space, let's discuss rotating objects in 3D space. Select *Monkey.jpg* and press *"R"* to view the rotation property. Like the position property you'll see that now has three properties instead of two: *X-rotation*, *Y-rotation* and *Z-rotation*.



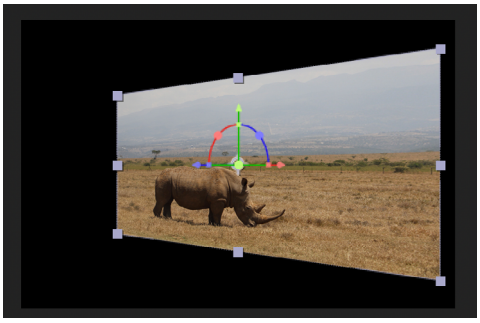
With *Monkey.jpg* selected changed the *X-rotation* to *315˚*. You can see that rotating in the X-rotation axis rotates the layer towards and away from the camera. Next select *Rhino.jpg* and set the *Y-Rotation* to *45˚*. You can see the *Rhino.jpg* rotates left to right, as if it were a top. So it rotates as if viewed from above.



Finally, select *Gazelle.jpg* and use the *Z-rotation*. You'll see that the image rotates towards the camera, similar to how the rotation property applied in previous labs.

Scale

Finally let's discuss scaling an object in 3D space. Highlight *Rhino.jpg* and use the *solo* toggle to isolate it. Press the *"S"* key to open up the scale properties and you'll see that like the position and rotation that you have three properties. You can scale the *x-axis*, *y-axis* and *z-axis*.



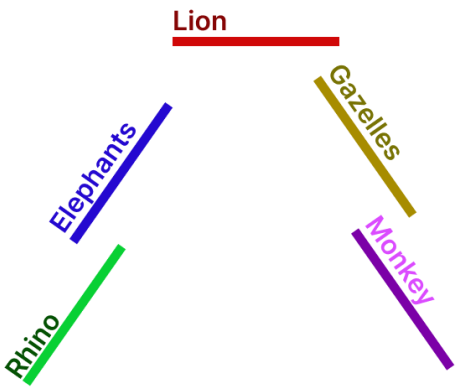
Uncheck the link icon so you can scale each dimension

independently of each other. If you scale the x-axis you'll see that rhino image width increases. If you scale the y-axis you'll see that the rhino image's height increases. This is similar to how scaling images and videos worked previously. Finally let's scale the z-axis or the depth of the image. You'll notice that changing this value has no effect on the image. That is because unlike formal 3D programs like Maya or Blender layers in Adobe After Effects have no depth. They can not be thicker or thinner. Instead layers in After Effects are more like playing cards. They stand with a width and height, but no real depth, even when scaled. It's important to understand that when manipulating layers in Adobe After Effects.

Turn off the *solo* toggle to see all three layers in 3D.

Layout

For our video intro we want our images to be positioned and rotated a certain way. You can view the image on the right to see how the images will be positioned and rotated towards the camera.



Add the remaining images and use the guide below to position and rotate images accordingly. You may want to use the top view to make sure your images are place accordingly.

Rhino.jpg
position: -315, 240, 0
rotation: 305˚

Monkey.jpg
position: 1050, 240, 415
rotaition: 55˚

Gazelle.jpg
position: 860, 240, 1150
rotation: 55˚

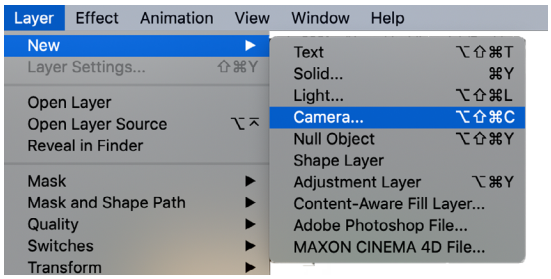
Elephant.jpg
position: -120, 240, 700
rotation: 305˚

Lion.jpg
position: 360, 240, 1800
rotation: 0˚

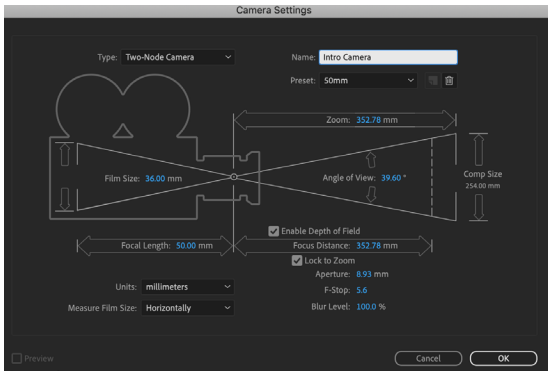


Cameras

So far we've learned how to place layers in Adobe After Effects in 3D space. Now let's learn how to build and animate cameras in After Effects. Let's revisit our analogy of building a diarama. Let's imagine that after we built a diarama we placed and moved a video camera to view our diarama. This is similar to what we will be doing with cameras in After Effect. We can build a camera, then use it to act as the perspective for our scene. Then if we want to we can animate the camera itself by changing it's position and rotation.



To build a camera in Adobe After Effects go to *Layer > New > Camera*. This will open a new dialog box that looks like the image on the right. This dialog box allows us to set the settings of the video camera we will build. We won't too much about the setting at the moment, but let's give our camera a name. This name will appear in the 3D View Popup menu and in our layers. Label the camera *"Intro camera"* and set the preset to *"50mm"*. Click the Ok button at the lower right hand corner.



You'll see that a new camera layer is added in your timeline with a small video camera icon to the left of the name. If you view the top perspective you'll see the camera represented as a triangle.



Camera Basics

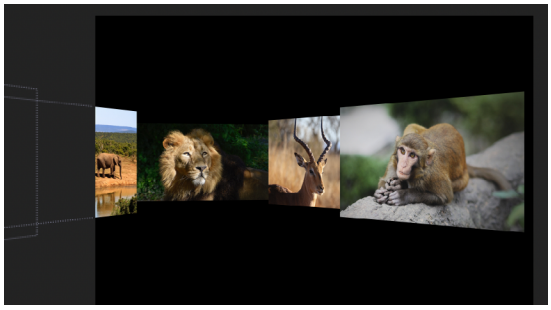
Set your 3D View Popup to your new camera. View how the scene looks through your camera. Cameras, like layers, can be placed and rotated in 3D space.

Select the camera and press the *"P"* key to gain access the position properties. Use the *z-position* to move the camera from *-1000* to *-2300*. We have now moved the camera farther away from the images and the images in the scene look smaller. If you view the top perspective you'll see that camera triangle is farther from the images than originally.



350˚Y-Rotation

In addition to changing the position of a camera we can also rotate a camera. Rotating the camera will rotate the view of everything in the scene. Select the camera in the layer and press the *"R"* key to access the rotation property. Set the *Y-Rotation* to *350˚*, this will rotate the camera viewing the scene to the left and all of the images will appear to have moved to right. Change the *Y-rotation* to *10˚*, this will rotate the camera viewing the scene to the right and the scene will appear to have shifted to the left.



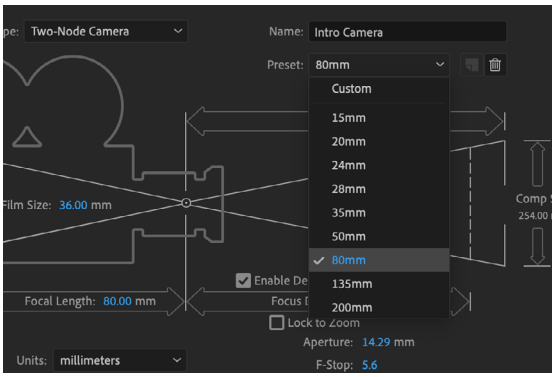
10˚Y-Rotation

Set the camera position to *200, 200, -2300* and reset the camera's *Y-Rotation* to *0*.

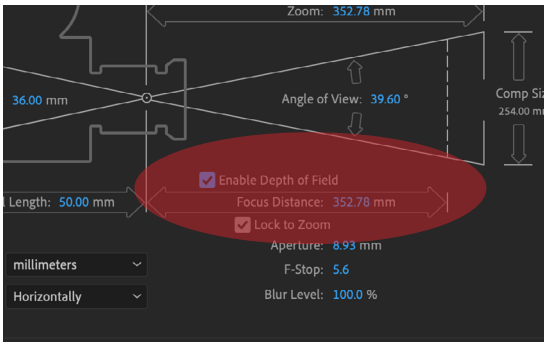
Camera Settings

In the real world cameras have lens. Lens are curved pieces of glass that bend light and focus the light on an item(s) in the scene. The length of the lens measured in millimeters is your focal length.

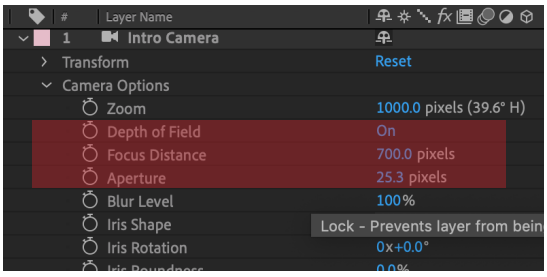
The higher the focal length the closer to the object you'll appear. A low focal length is good for capturing the entire environment of your scene, a higher focal length is good for focusing and framing a specific object.



Double click on the *Intro Camera* in your timeline to return to the camera settings. Change the *Preset* from *50mm* to *80mm* and hit okay. You'll notice that allow the camera position has not changed that the images in the scene feel closer to you. Double click the *Intro Camera* icon again and change the *Preset* to *35mm* and hit okay. You'll see that the view of the scene now appears further away, even though the camera's position has not changed. Double click on the *Intro Camera* and reset the *Preset* to *50mm*.



Return to camera settings and click the checkbox for *Enable Depth of Field*. *Depth of Field* is a photography term for whether an object in the distance is within focus inside the camera. For instance if a camera is focusing on a person, the mountains in the background tend to a be a bit blurry due to *depth of field*. By default cameras in Adobe After Effects have no *depth of field*, everything is a 100% clear. We may want to enable depth of field to have our virtual camera work more like cameras in the real world.



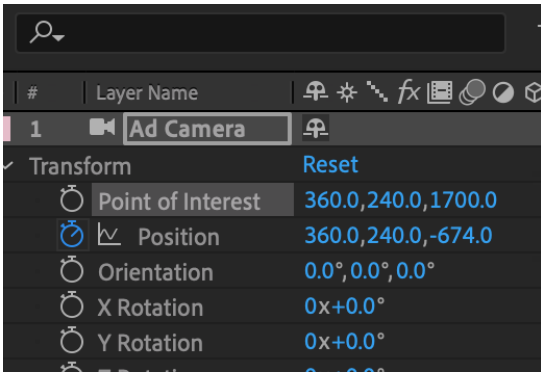
In the timeline open the *Camera Setting* scrolldown for *Intro Camera* and find the property called *Focal Distance*. This distance represents that any items within this range will be in focus. Anything beyond this range will be out of focus and become blurry. At the moment our *Focal Depth* is *1000px*, let's lower that to *700px*. You'll notice that the Rhino and Monkey images are clear while the *Lion.jpg* photo in the back is a bit out of focus. This makes our virtual camera act more similar to a real life camera.

Animate a Camera

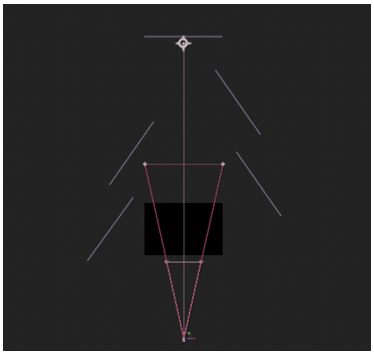
Now that we our camera setup let animate moving through our images. We want to animate the camera moving forward going through the images until it reaches the *Lion* image in the back.

Move the time marker to the beginning of the timeline. Keyframe the *Z-position* for the *Intro Camera* at *-2300*. Move the time marker to 3 seconds and set the *Z-position* to *950*.

Use the *Preview* panel to view the animation, you can from the Top prespective the camera moving through scene. You might notice that as the camera is moving it suddenly dips to black. If you view the *Top* perspective you will see that at the end of the animation that camera appears to flip from pointing upwards to pointing downwards, why is that? The answer is because of something called the *Point of Interest*. The *Point of Interest* is a spot that that our virtual camera focuses on. If you move or rotate the camera it moves or rotates in relation to that *Point of Interest*. Think of the point of interest a bit like the anchor point of a layer. The issue here is the camera is moving beyond the point of interest, which is causing the camera to flip to remain looking at the it.



The solution to this issue to change the *Point of Interest*. Go to timeline and open the properties of the *Intro Camera*. At the top with be the point of interest coordinate. Let's adjust the *z-position* of this and move it be right in front of our lion photo. That way the camera will always move and rotates towards that point. Change the *z-position* from *0* to *1700*. In the *Top* perspective the point of interest is shown as a crosshair icon.



Preview the animation again and it should appear correctly with the camera stopping right before the lion.

2D Objects in a 3D World

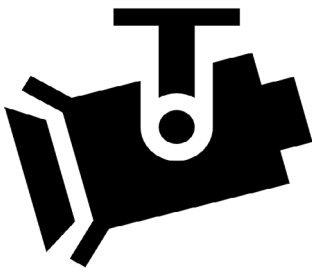
This nature documentary is being funded by the World Wildlife Fund. They would like their logo to go in the lower right hand corner of the video intro. They would like the logo to always be visible during the graphic, meaning the virtual camera should not move past it like it does for the animal images.



Drag and drop *Logo.png* into the timeline and use the *Selection* tool to place it at the lower right hand corner. Set the *opacity* to *25%*. Make sure to NOT make the logo a *3D* layer in *Toggle Switches*.

Preview the animation again and you will see that as the camera moves the logo stays locked. Non 3D layers will not move in 3D space and will stay locked to the active camera as they did in previous lab.

Lighting



In the world of theater and movie-making lighting has alway played a very important role. After all, lighting allows you to thesee the world, no lights means, no world!

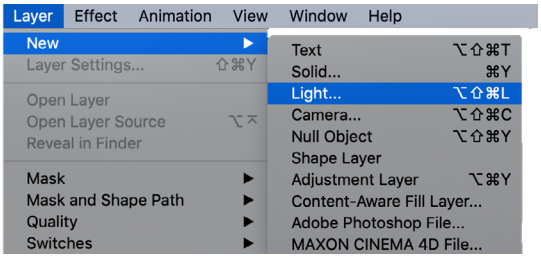
Adobe After Effects allows you to create your own basic lighting

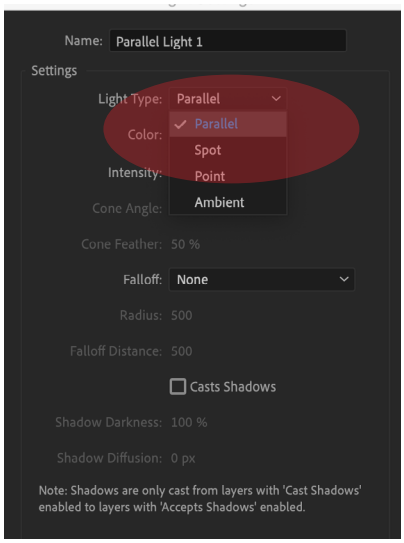
and allows you to control elements of that light including intensity, color and angle.

Begin by creating a new composition with the following settings

Name: Lighting
Dimensions: 720 x 480px
Frame Rate: 30
Pixel Shape: Square
Duration: 00:00:08:00

Drag and drop the 3D composition into the new composition. To create a new light in Adobe After Effects go to *Layer > New > Light* to open a new dialog box. From the dialog box select *Light Type* and choose *Parallel*, we will discuss settings later. Hit okay to create a new light layer in your timeline. You can tell if something is a light as it will have a lightbulb icon to the left of the name,





You may notice adding the light has not affected the composition. That is because lights in Adobe After Effects only affect 3D layers. Turn on the 3D toggle for the 3D layer and you will see the light begin to affect the scene.

Lighting Settings

Double click on the lightbulb icon to open the light settings for the light. There are four types of lights that can be used in Adobe After Effects: *ambient*, *spot*, *point* and *parallel*. Here is the basic breakdown for each type.

Parallel lights: mimic light being delivered by the sun. All objects in the scene will get the same amount of light regardless of the placement in the scene so placement in the scene is irrelevant. Parallel lights do not shine light in a specific direction or range.

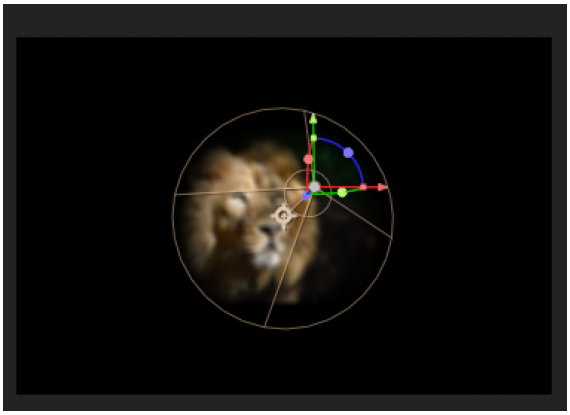
Point Lights: Mimic sources that emit light like a candle. Light is released in all directions from the origin point of the point light. Point lights only affect objects within the range of the point light.

Spot Light: Spot lights are similar to spot lights found in theatres and movie sets. They are lights that are focused on a specific size and direction.

Ambient Lights: Ambient light is similar to Parallel lights in that it affects every object in the scene. Light parallel light there the light does not shine in a range or direction.

For this project we want to focus the attention of the users to the center of the video, thus, we need a light that can rotated, or has direction, and does not affect everything equally, or has what is

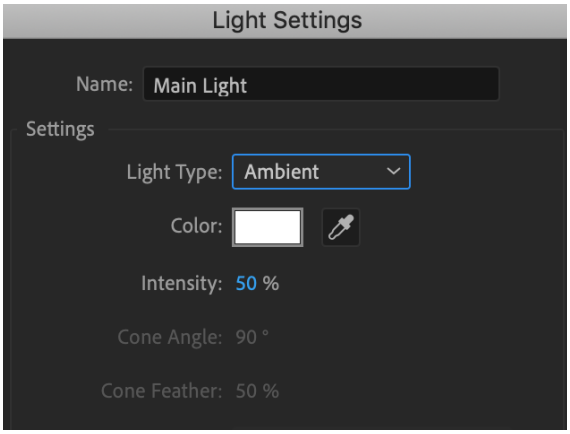
referred as falloff. Let's change *Light Type* to *Spot*. In the *Name* area rename the light to *"Main Light*". Hit "Ok" and you can see you scene's lighting changes dramatically. The spot light is shining on the center, the Lion in this case and the rest of the scene is in darkness.



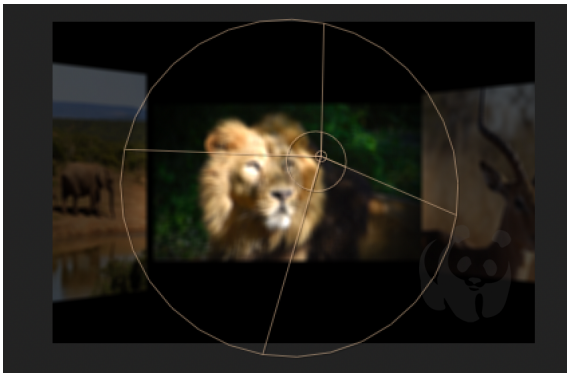
Spot lights have a cone, or the size of the light shinning on a surface. To increase what we can see in the scene go the *Main Light* in the timeline and find the *Cone Angle* to *120˚*. The spot light surface gets larger and we can see more of the scene. Change the *Cone Angle* again to *60˚* and we can see the light become smaller and more focused. Let's set the light cone angle to about *90˚*.

Add Additional Lights

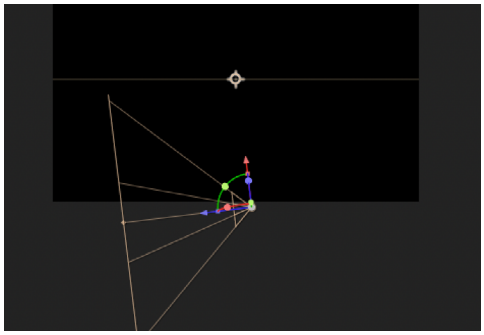
Although we used the spot light to focus the audience's attention to the center, we are losing some of the images and other content around it. Let's help fill in the gaps by using an ambient light. The ambient light will affect the whole scene and is not dependent on it's location.



Go to *Layer > New > Light* and choose *Ambient* for *Light Type*. How bright the light is is the *Intensity* option, Set it to *30%*. Hit *Ok* and you will see that additional lights are helping to light other areas of the scene, but they do not overpower the main spot light.

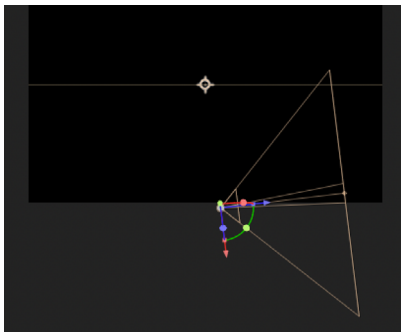


Add an Animated Light

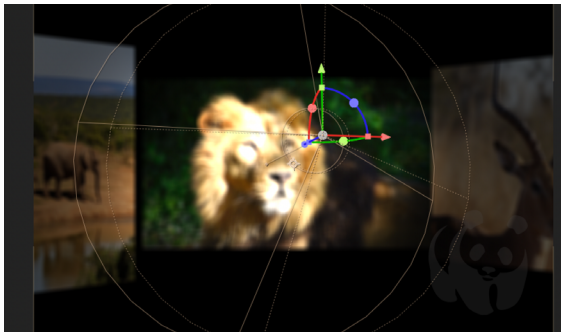


After the camera moves through the scene it will hold on the photo of the lion for a few seconds. While waiting it would be fun to create a shine effect, or a spotlight that moves across the scene. We can do that by animating a light in Adobe After Effects.

Go to *Layer > New > Light* and choose *Spot*. Set the *Intensity* to *50%* and name the light "*Shine,*" then hit *Ok*. The result is having two bright spot lights pointing towards the scene, which is making the scene to bright. To solve this issues we can rotate the spot light to point the left side of the screen, rather than on the canvas itself.



Select the *Shine* light and hit *"R"*. Use the *Y-Rotation* and set it to *-90˚*. If you use the *Top* perspective you can see the light being rotated to the left. Go to the 3 second mark in the timeline and set a keyframe for the *Y-Position*. At the 4 second mark change the *Y-Rotation* to *90˚*. The result will be *Shine* light we rotate across the page, causing a quick shine effect on the video.



Use the *Preview* to view the motion graphic as a whole.

Animated Type

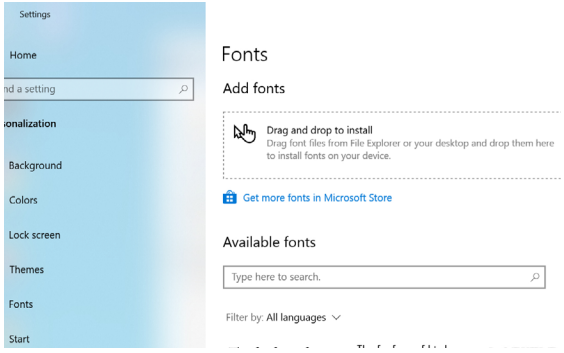
Phew, we covered a lot when it comes to 3D, cameras and lights. The end result is a cool video of a camera moving through a 3D space with the wild African animals in it.

The last thing we need to create our title text to finish this intro. Create a new composition called *Type* with the following settings:

Name: Type
Dimensions: 720 x 480px
Frame Rate: 30
Pixel Shape: Square
Duration: 00:00:08:00

Nest the *Lighting* composition into *Type*.

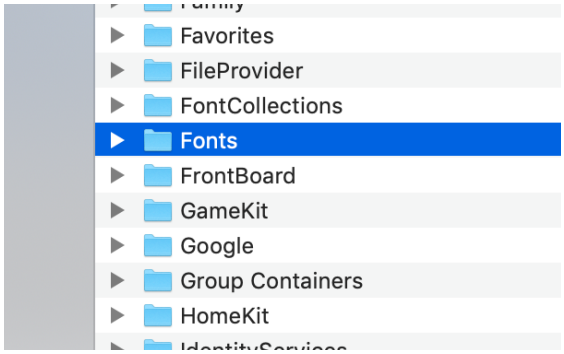
Install Fonts



Our clients want to use a specific typeface for this project called Trade Winds. In order to use this typeface in our project we must first install the font fille into our computer. The computer needs the font file in order to create the typeface.

How to install fonts depends on your operating system. If you are using a computer running Windows go to the *Windows*

button at the lower left, go to *Settings > Personalization > Fonts*. Then you can drag and drop the font file into the *Add Fonts* areas to install the font.



If you using a Mac computer begin by going to the *Desktop*. Go to *Go* menu and hold the *option* key. This will allow you access to the *Library* section. In *Library* find the folder marked *"Fonts"*. Then drag and drop the font file into the *Fonts* folder to install it.

Install *Trade Winds* for your computer before moving on to the next section.

Stylize Type

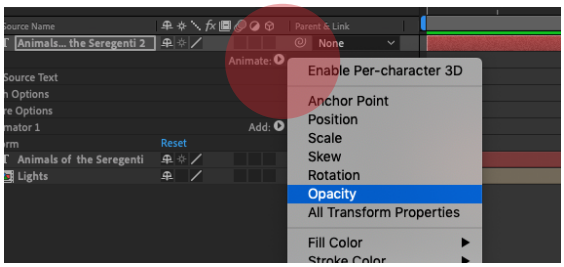
Use the *Type Tool* in the menu and write the phrase *"Animals of the Seregenti."* Use the *Character* and *Paragraph* panels to format the title:

Typeface: Trade Winds
Font Size: 76px
Leading: 55px
Alignment: Center



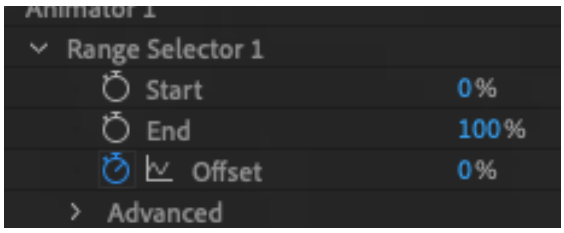
Text Animators

In the past we've animated a text layer using layer properties such as opacity or position. These properties effect the entire block of text, whether it is one word or a whole paragraph. What we will do in this lab is animate each character within the text block individually. To do this we will use what are referred to as *Text Animators*.



Select your title and in the timeline click on the option marked Animator to add a *Text Animator* to your text. In the popup menu choose the *Opacity* option. This will have opacity effect each letter instead of the text block as a whole.

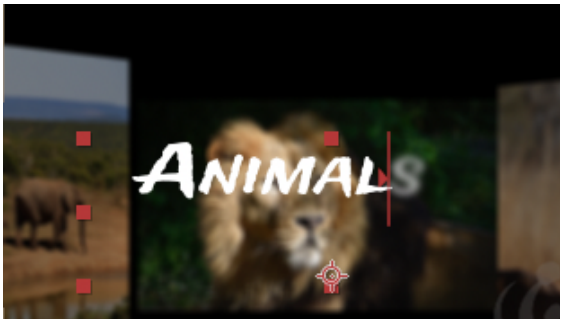
Set the time marker to the 6 second mark. In Animator find the option marked *Offset*. The offset option intrepolates between the default of the text settings of the text and the offsets to the text applied by the *Text Animator*. Set the offset option to *0%*. Then move the time marker to the 5 second mark and change the value of the *Offse*t to *100%*. This is animating between the *0%* or the Animator value and *100%* or the original text value. Use the *Preview* panel to view the Intro, you should see that each letter fades up as opposed to the text block as a whole.





Now that we have one animated property, let's add a second property. Find the Animator in the timeline and click on the *Add* button. This will open up a list of additional properties we can

animate as part of the text animation. Choose the *Blur* option, this will allow us to animate each letter starting blurry and becoming clear over time. Set the *Blur* value to *20px*. Preview the animation and you will see each letter starting blurry and becoming clear as it fades up.



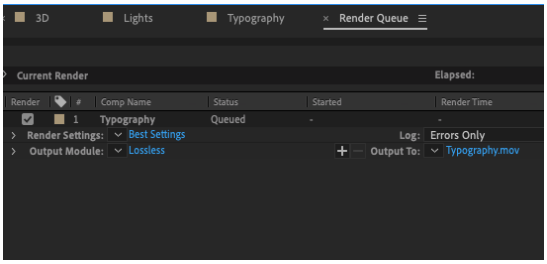
Finally, let's add a *Rotation* to the *Text Animator*. This will have the text start slightly rotated and then rotate into place while also fading up. Select the Animator and choose the *Add* button to add another property to animate. Choose *Rotation*. Set the *Rotation* value to *45˚* to have it start rotated forward. Preview the Intro sequence and you will see each letter rotate while also fading up from being blurry.



Finally, let's add a drop shadow to the text layer so it stands out from the background video. Select the text and go to *Effect > Perspective > Drop Shadow*.

Export

Great job! Now let's export the final project to show the client. With the *Type* composition open in the timeline go to *Composition > Add to Render Queue*.



In the *Render Queue* look for *Comp Name* make sure it is set to *Type*.

Under *Render Settings* make sure it is set to *"Best Settings".*

Under *Output Module* make sure it is set to *"Lossless"* make sure the format is set to either Quicktime or AVI.

Finall, use the *Output To* option save the animation to your Desktop with the label *lastname-3D.*

Submission

Upload your exported project video to your YouTube account. Publish the video and you will copy and paste the video URL to the submission page for the lab.

In addition to the final video url please upload the *lastname-intro.aep* file to Canvas for grading. Please do NOT submit all of the files or videos as those are large files, instead just upload the *.aep* project file for grading purposes.